



**उत्पाद मानुयल
आई एस 269:2015 के अनुसार
साधारण पोर्टलैंड सीमेंट के लिए**

दस्तावेज संख्या- पी एम/आई एस 269/3/अक्टूबर 2023

भारतीय मानक ब्यूरो (अनुरूपता मूल्यांकन) विनियम की स्कीम-1 के तहत यह उत्पाद मानुयल प्रमाणीकरण के प्रचलन में रीति और पारदर्शिता के सुसंगत सुनिश्चित करने के लिए सभी क्षेत्रीय/शाखा कार्यालयों एवं लाइसेंस धारियों द्वारा संदर्भ सामग्री के रूप में उपयोग किया जाएगा। बीआईएस लाइसेंस/प्रमाण पत्र प्राप्त करने के इच्छुक भावी आवेदकों द्वारा भी इस दस्तावेज का उपयोग किया जा सकता है।

**PRODUCT MANUAL FOR
ORDINARY PORTLAND CEMENT
ACCORDING TO IS 269: 2015**

Document No.- PM/ IS 269/3/Oct 2023

This Product Manual shall be used as reference material by all Regional/Branch Offices & licensees to ensure coherence of practice and transparency in operation of certification under Scheme-I of Bureau of Indian Standards (Conformity Assessment) Regulations, 2018 for various products. The document may also be used by prospective applicants desirous of obtaining BIS certification licence/certificate

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PRODUCT MANUAL FOR ORDINARY PORTLAND CEMENT ACCORDING TO IS 269: 2015

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1.	Product	:	IS 269 : 2015
	Title	:	Ordinary Portland Cement
	No. of Amendments	:	1
2.	Sampling Guidelines:		
a)	Raw material	:	Performance improver (if added) – as per clause 5.1.1 and Table 1 of IS 269
b)	Grouping guidelines	:	Please refer ANNEX – A
c)	Sample Size	:	8 kg for physical test 1 kg for chemical test.
3.	List of Test Equipment	:	Please refer ANNEX – B
4.	Scheme of Inspection and Testing	:	Please refer ANNEX – C
5.	Possible tests in a day :		
	(i) Insoluble Residue (ii) Loss of ignition (iii) Fineness (iv) Setting time		
6.	Scope of the Licence :		
	Licence is granted to use Standard Mark as per IS 269 : 2015 with the following scope:		
	Name of the product	Ordinary Portland Cement	
	Grade	OPC 33/ OPC 43/ OPC 53/ OPC 43S/ OPC 53S	
	Any other aspect required as per the Standard	Transverse strength test is optional test as per agreement between purchaser and manufacturer	

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ANNEX A

Grouping Guidelines

1. Ordinary Portland Cement as per IS 269:2015 are classified as follows:

Grade: OPC 33, OPC 43, OPC 53, OPC 43S & OPC 53S

2. Considering the above, grouping guidelines as given below shall be followed for GoL/ CSoL:

Each grade of Ordinary Portland Cement shall be tested for all requirements to cover that particular grade of cement.

3. When a manufacturer intends to use a performance improver, the manufacturer shall submit the following for approval by BIS:

- a) Declaration of performance improver intended to use (type & percentage)
- b) Report of conformity of performance improver through supplier test certificate/in-house test report; and conformity of cement (produced with intended performance improver) through test report from BIS approved third party laboratory.
- c) Cement bag design artwork indicating the marking of performance improver

Firm shall declare the type and percentage of performance improver(s) being added to manufacture the cement and the same shall be recorded in the factory test record. (see form No 6 & 8 of SIT).

4. The Firm shall declare the varieties of Ordinary Portland Cement they intend to cover in the Licence. The Scope of Licence may be restricted based on the Manufacturing and Testing capabilities of the Manufacturer.
5. During the operation of the Licence, BO shall ensure that all the varieties covered in the Licence are tested in rotation, to the extent possible.

ANNEX B**List of Test Equipment***Major test equipment required to test as per the Indian Standard*

SI No.	Tests used in with Clause Reference	Test equipment
1	Fineness Clause 7 Table 3	Blaine's apparatus variable flow type
		Stop watch with start-stop mechanism
		Mercury for calibration
		Balance, Standard weights
		Standard Cement
		Manometer liquid (di-butyl phthalate or light mineral oil.)
		Mercury of reagent grade or better, Pyknometer
		Circular discs of filter paper of medium porosity (mean pore diameter 7 μ).
		Le-Chatelier's flask
		Constant temperature water bath to maintain temp. within ± 0.1 ° C
2	Soundness by Autoclave Clause 7 Table 3	Auto clamp machine with thermostatic control to maintain pressure of 2.1 MPa for 3 hrs, pressure to be attained within 1-1 ¼ hrs; L-Shape thermometer LC 1 °C
		Pressure gauge 0-42 kg/cm ² LC = 0.4 kg/cm ²
		Humidity chamber with temperature & RH control 27 $\pm$ 2 °C, RH 90 to 100 %
		Standard bar 308 mm, max
		Bar moulds 25 x 25 x 282 mm
		Length comparator with dial gauge
		Mineral oil for covering moulds
3	Soundness by Le-chatelier Method Clause 7 Table 3	Le-Chatelier's water bath preferably with thermostatic control raising temperature from 27 $\pm$ 2 °C to boiling in 27 $\pm$ 3 minutes
		Le-Chatelier's moulds with weights and cover glasses minimum 8 nos.
		Humidity chamber with temperature & RH control 27 $\pm$ 2 °C, RH 90 to 100 %
		Steel scale 12" (304.8 mm)

4	Setting time Clause 7 Table 3	Vicat apparatus
		Needle for Consistency, IST& FST testing
		Moulds
		Stop Watch
		Balance - 1000g $\pm$ 0.1g and Standard Weights 1mg to 500 gm
		Gauging trowel of weight 210 $\pm$ 10 g
5	Compressive Strength Clause 7 Table 3	Vibration machine with timer & cube mould fitting assembly 12000 $\pm$ 400 vibration per min.
		Compressive Strength machine
		Poking Rod , Petroleum Jelly
		Proving ring with all accessories suitable for calibration of CST machine
		Tachometer
		Cube Moulds 70.6 $\times$ 70.6 mm, Poking rod
		Gauging trowel (210 $\pm$ 10 g) gauging plate, stainless steel(non-perforated)
		Standard sand grade 1, 2 and 3 (as per IS 650)
		Curing tank of appropriate size with water circulation arrangement
		Graduated glass cylinders 150 to 200 ml
		Humidity chamber with temperature & RH Control 27 $\pm$ 2° C, RH 90 to 100 %
6	Transverse Strength Test Clause 7, Table 3	Moulds
		Planetary Mixer, Standard Sand
		Jolting Apparatus, Scraper, Demolding device as per IS 4031 (Part 8)
7	IS 269 : 2015	General equipments for Cement testing
a)	To control humidity & temperature in lab	1. Humidity chamber with temperature & RH control 27 $\pm$ 2 °C, RH 90 to 100 % 2. Suitable arrangement to demonstrate maintenance of temp. of 27 $\pm$ 2 ° C & RH 65 $\pm$ 5% constantly
b)	For cement Sampling	Mixing trays –adequate size including trays of 24 partitions for keeping hourly samples
c)	To control the residue of cement	Sieves of size (300, 212, 150, 90, 75 & 45 μ)
d)	To measure temperature	Thermometers

e)	Lab ball mill (motorized)	To grind the clinker, slag & gypsum sample in lab ball mill for testing
f)	To weigh the material	1. Platform type balance 2. Electrical balance 3 Weight box with weights (1 mg - 500 g)
9	General test equipments for chemical testing Clause 6 , Table 2 of IS 269 : 2015	
		Muffle Furnace with thermostatic control, Range 0 – 1200° C
		Oven with thermostatic control 0-300° C
		Heater and hot plate
		Distillation Assembly
		Crucible: Platinum or Porcelain / silica
		Filter paper (No- 1, 40, 41, 42)
		Desiccators with cover & Desiccant
		Water bath
		pH meter/paper
		Glassware - volumetric flask -0-250 ml, beaker 0-250 ml, measuring cylinder 0-50,100,500, 1000 ml, burette 0-25/50 ml, conical flasks- 0-250 ml, pipette 0-5,10, 25, 50 ml
		All chemicals required for complete chemical analysis of cement
		Tongs including platinum tipped tong
		Wire gauge with asbestos sheet at the middle
		Washing bottle
		Mortar mixer- 4.75 l
		Glass thermometer
		All required chemicals as per IS 4032 Ordinary portland cement and Portland clinker testing.
10	Testing arrangement for performance improver as per Clause 5.1.1 and Table 1 of IS 269 : 2015	

The above list is indicative only and may not be treated as exhaustive.

ANNEX C**SCHEME OF INSPECTION AND TESTING**

1. LABORATORY - A laboratory shall be maintained which shall be suitably equipped (as per the requirement given in column 2 of Table 1) and staffed, where different tests given in the specification shall be carried out in accordance with the methods given in the specification.

1.1 The manufacturer shall prepare a calibration plan for the test equipments. The following equipments shall be calibrated at a frequency shown against each and records kept.

Sl. No.	TEST EQUIPMENT	FREQUENCY OF CALIBRATION
1.	Blaine's apparatus	Daily with licensee's own Standard cement sample and monthly with standard cement samples supplied by NCCBM.
2.	Compressive strength Testing machine	Once in a month with Licensee's own Proving Ring and the Proving Ring shall be Calibrated once in two years from a NPL/NABL Accredited Calibrating body or NPL or NPL accredited Proving Ring manufacturer.
3.	Autoclave pressure gauge	Once in a month by licensee's own dead weight pressure gauge tester OR once in six months from accredited calibrating body or NPL/NABL accredited manufacturer of such gauges.
4.	Vibration machine	Once in a month by licensee's own Tachometer. The tachometer shall be calibrated once in a year from NPL/NABL accredited outside agency.
5.	Dead weight pressure gauge Tester (if available)	Once in four years from NABL accredited Tester (if available) Lab or OEM (original Equipment manufacturer) having NPL/NABL accredited calibrator.

2. TEST RECORDS – The manufacturer shall maintain test records in various formats, Form 1 to Form 18 for the tests carried out to establish conformity.

3. LABELLING AND MARKING – Labeling and marking shall be as given below:

3.1 STANDARD MARK - The Standard Mark, as specified by BIS, shall be printed or stenciled on each bag or drum of Ordinary Portland Cement or on the label applied to it, provided the material in each bag or package to which the mark thus applied conforms to the specification. The size of the Standard Mark shall be either **160 x 120 mm** or **80 x 60 mm** for packing in quantity of 50 kg and above. For other packing of lower quantity, a photographic reduction is permitted.

3.2 MARKING - As per the requirements of IS 269: 2015.

3.2.1 In addition to above, following marking shall also be marked:

- Name of original manufacturer of cement with BIS licence number in case of repacking unit.
- Licence no. CM/L-_____ and details of BIS website, i.e. "For details of BIS certification please visit www.bis.gov.in".
- Any other marking required under provisions of Legal Metrology Act, 2009 and Legal Metrology (Packaged Commodities) Rules, 2011 framed thereunder.

3.2.2 All the information including BIS Standard Mark except Manufacturers Registered Trade Mark shall be applied on each bag in **BLACK COLOUR**.

Note :

1. For each calendar year, the first week shall be counted as 7 days from 1st of January and subsequent weeks numbered serially accordingly. The bags shall be marked as W 01/MM/YY..... W 51/MM/YY..... etc.
2. Label mentioned at 3.1 and 3.2 above shall be attached to the seal of the container. The seal shall be of such a design that it shall automatically get destroyed on opening.
3. The colour of the bag and background colours should be in contrast to the colour of the Standard Mark and the details so that the markings are conspicuous.

4. CONTROL UNIT –

4.1 For manufacturing units of Ordinary Portland Cement: The tests, as indicated in Table 1 attached and at the levels of control specified therein, shall be carried out on the whole production of the factory which is covered by this scheme and appropriate records maintained in accordance with clause 2 above.

4.2 For packing of Ordinary Portland Cement at bulk cement terminal: The tests, as indicated in Table 2 attached and at the levels of control specified therein, shall be carried out on the whole packing of Ordinary Portland Cement and appropriate records maintained in accordance with clause 2 above.

4.2.1 For bulk packing units as per clause 4.2, all cement of one consignment received shall constitute one batch.

4.2.2 Batch mixing may be permitted for packing units, which are extended packing terminals of the same cement manufacturer (licensee) subject to packing units obtaining test certificates from the manufacturer and keeping proper records. If the cement is received from different units of the same manufacturer (different licensees) batch mixing of cement is not permitted. The Batch integrity shall be ensured at all stages of packing and the packer shall maintain appropriate controls and checks to ensure that there is no chance of mix up of different batches. Adequate care shall be taken to avoid spoilage during handling, packing and storage.

4.2.3 If bulk packing unit is instructed by BIS for suspension of licence due to the failure of the samples, such instruction will automatically apply to the original manufacturer of the cement, as per relevant suspension of licence guidelines. An undertaking to this effect shall be obtained from the bulk packers and the original cement manufacturer.

4.2.4 Test Certificate of each original batch of cement shall be obtained from the supplier and test results recorded. On the basis of tests and inspection, the decision regarding conformity or otherwise of the consignment/batch to a given requirement shall be taken.

4.3 WEIGHMENT – One filled bag from each nozzle shall be taken at random twice in each shift of operation and weight checked in case of electronic packers with recorders. In all other cases one filled bag from each nozzle shall be checked once in two hours. The records shall be maintained in Form 1. The bag shall be so chosen for weighment such that bags from each nozzle are taken for weighment. The weighing and packing machines shall be adjusted as and when necessary in such a way that net quantity of each bag shall be in accordance with the

tolerances given in Annex B and clause 10.1.1 of IS 269 : 2015. Such adjustments for each nozzle shall be recorded in Form 1 under remarks column.

4.3.1 For packing of Ordinary Portland Cement in bulk cement terminal weighment of hourly check of mass of drums also shall be done in addition to weighment of bags mentioned in para 4.3 above. The records of weighments shall be maintained in Forms 10 and 12.

4.4 RAW MATERIALS

4.4.1 Routine analysis of various raw materials used in the manufacture of Ordinary Portland Cement shall be made at intervals of a month or whenever there is a change in the source/mine area stratification whichever is earlier and appropriate records of the analysis and of the Physical composition of the mixtures shall be maintained in Form 2. This analysis is not applicable for Packing Units of Ordinary Portland Cement at bulk cement terminal.

4.5 PERFORMANCE IMPROVER – Manufacturer shall declare performance improver and its percentage added in the ordinary portland cement. The details of such performance improver added to cement and its percentage shall be printed on bags/package of cement.

4.6. PACKING - The Cement shall be packed in bags as specified in clause 10 of IS 269: 2015. A test certificate either from the manufacturer or from any recognized testing laboratory shall be received along with each consignment of bags. Alternatively the samples of bags from each consignment shall be tested by the cement manufacturer either in his own laboratory or any other BIS recognized laboratory before they are used for packing cement. No testing would be necessary if the bags carry BIS Certification Mark. The bag shall be in good condition at the time of packing.

5. LEVELS OF CONTROL - The tests as indicated in column 1 of Table 1 and the levels of control in column 3 of Table 1, shall be carried out on the whole production of the factory which is covered by this plan and appropriate records maintained in accordance with clause 2 above.

5.1. PRODUCTION DATA - The licensee shall send to BIS a statement of quantity produced, marked and exported by him and the value thereof at the end of each quarter of the operative period as per the enclosed proforma and shall also submit these details to BIS at the end of the operative year duly authenticated by a Chartered Accountant.

6. REJECTIONS – Disposal of non-conforming product shall be done in such a way so as to ensure that there is no violation of provisions of BIS Act, 2016.

Table 1 – Levels of Control (Grinding/Packing Unit)

(1)				(2)	(3)			
TEST DETAILS				Test equipment requirement R: required (or) S: Sub-contracting permitted	RECOMMENDED LEVELS OF CONTROL			
Clause	Requirement	Test Methods Clause Reference			Number of sample	Frequency		Remark
						Cement Grinding/Blending	Cement Packing	
5.1 and 5.1.1, Table 1	Performance improver	5.1.1	IS 269	S	One	One sample shall be tested from each supplier once in a month/whenever there is change of source of supply of material as per relevant Indian Standard as given in Table 1 of IS 269: 2015. No further testing required if accompanied with test certificate or ISI marked.		
6 Table 2	Chemical Requirement							
i)	LSF	6	IS 269	R	One	Daily Composite sample	Weekly composite sample	-
ii)	AF Ratio	9	IS 269	R	One	Daily Composite sample	Weekly composite sample	-
iii)	Insoluble residue	6	IS 269 IS 4032	R	One	Daily Composite sample	Weekly composite sample	-
iv)	Magnesia	6	IS 269 IS 4032	R	One	Daily Composite sample	Weekly composite sample	-
v)	Total Sulphur content calculated as sulphuric anhydride (SO ₃)	6	IS 269 IS 4032	R	One	Daily Composite sample	Weekly composite sample	-
vi)	Loss on ignition	6	IS 269 IS 4032	R	One	Daily Composite sample	Weekly composite sample	-
vii)	Chloride Content	6	IS 269 IS 4032	R	One	-	Weekly composite sample	This test shall also be carried out whenever there is any change in source of any raw material.

viii)	Alkali Content	6	-	S	One	-	-	Pl see note under Table 2 of IS 269: 2015.
7 Table 3 Physical Requirement								
i)	Fineness	7	IS 269 IS 4031 (Part 2)	R	One	1. Every alternate hourly from each mill separately. 2. Daily Composite sample	Daily Composite sample	-
ii)	Soundness (Le-Chatelier method and Autoclave method)	7	IS 269 IS 4031(Part 3)	R	One	Daily Composite sample	Daily Composite sample	-
iii)	Setting Time	7	IS 269 IS 4031 (Part 5)	R	One	One sample per shift (Composite sample)	Daily Composite sample	-
iv)	Compressive strength	7	IS 269 IS 4031 (Part 6)	R	One	Daily Composite sample	Daily Composite sample	-
v)	Transverse strength (Optional)	7	IS 269 IS 4031(Part 8)	S	One	Weekly composite sample	Weekly composite sample	-

NOTES –

- Composite sample shall be made out of hourly samples for the required period (Pl see IS 3535 Methods of sampling hydraulic cements).
If cement is manufactured using same proportion of raw materials from more than one cement mill, sample from each mill shall be tested for fineness as per the above table.
For all other parameters composite samples from all the mills shall be tested.
If cement is manufactured using different proportion of raw materials from more than one cement mill, sample from each mill shall be tested for all requirements as per the above table.
- For manufacturing units where there is no packing silo and cement is packed directly from cement grinding, the frequency of tests specified for cement grinding stage would apply for the various tests to be carried out on samples taken from cement mill spouts along with weekly chloride content test.

Table 2 Level of Control (Bulk Packing Unit)

(1)				(2)	(3)		
TEST DETAILS				Test equipment requirement R: required (or) S: Sub-contracting permitted	RECOMMENDED LEVELS OF CONTROL		
Clause	Requirement	Test Methods Clause Reference			Number of samples	Frequency	Remarks
6 , Table 2 (iii)	Insoluble Residue	6	IS 269 IS 4032	R	One	Each batch	To be tested in laboratory at bulk terminal packing unit.
6, Table 2 (vi)	Loss on Ignition	6	IS 269 IS 4032	R	One	Each batch	
7, Table 2 (i)	Fineness	7	IS 269 IS 4031(Part 2)	S	One	Each batch	-
7, Table 2 (ii)	Soundness	7	IS 269 IS 4031(Part 3)	S	One	Each batch	
7, Table 2 (iii)	Setting Time	7	IS 269 IS 4031(Part 5)	S	One	Each batch	
7, Table 2 (iv)	Compressive strength	7	IS 269 IS 4031(Part 6)	S	One	Each batch	

Form No. 1

FORMAT FOR MAINTENANCE OF TEST RECORDS WEIGHMENT CONTROL AT PACKING STAGE

Date	Shift	Time (Hourly)	No. of Bags	Net mass of bags from nozzles						Remark.

Form No. 2

RAW MATERIAL TESTING

Date of receipt of material	Date of testing	Name of Material	Source of supply and consignment No.	Details of analysis for specified requirements

Form 3

PRODUCTION DATA

(POST GRINDING DETAILS OF PRODUCTION ACCEPTED AND REJECTED FOR STANDARD MARK)

Shift	Quantity	Passed for Standard mark	Rejected	Remark

Form No. 4

CLINKER CHEMICAL COMPOSITION (DAILY COMPOSITE SAMPLE)

Date of manufacture	Total loss on ignition	Insoluble residue	Alkali content as Na ₂ O @	C ₃ S	C ₃ A	C ₃ S+ C ₂ S	SO ₃	MgO	Chloride	Free lime	LSF	Alumina factor	Sample Pass/Fail	Remarks

@ Alkali content test to be conducted as and when desired by Purchaser

Form No. 5

CLINKER GROUND WITH GYPSUM (DAILY COMPOSITE SAMPLE)

Date of grinding	Fineness	Soundness		Setting time		Compressive strength			Sample Pass/Fail	Remark
		Le-Chatelier	Autoclave	Initial	Final	3 days	7 days	28 days		

Form 6

ORDINARY PORTLAND CEMENT (GRINDING/ BLENDING) (Daily/Weekly Composite sample)

Date of grinding	Type & % of performance improver	LSF	AF	IR	MgO	Total Sulphur calculated as SO ₃	LOI	Chloride Content	Alkali content @	Fineness	Soundness (Le-chatelier & Autoclave)	Setting Time Initial & Final	Compressive strength	Transverse Strength	Sample Pass/Fail	Action taken if sample fails

@ Alkali content test to be conducted as and when desired by Purchaser

Form No 7

ORDINARY PORTLAND CEMENT GRINDING (For Alternate Hourly Sample)

Date of grinding	Time	Fineness	Setting Time Initial & Final	Sample Pass/Fail	Mode of disposal or action taken if sample fails

Form No 8

ORDINARY PORTLAND CEMENT PACKING STAGE (Daily/Weekly Composite Sample)

Date of Packing	Type & % of performance improver	LSF	AF	IR	MgO	Total Sulphur calculated as SO ₃	LOI	Chloride Content	Alkali content @	Fineness	Soundness (Le-chatelier & Autoclave)	Setting Time Initial & Final	Compressive strength	Transverse Strength	Sample Pass/Fail	Action taken if sample fails

@ Alkali content test to be conducted as and when desired by Purchaser

Form No 9

CALIBRATION

Sl. No	Date of calibration	Result of Calibration (Test records indicating details of standard values and observed values for each equipment to be kept in proforma for which various columns be devised; as required)	Name of equipment Action taken if equipment found defective	Sl.No. (If any) & Remarks

Note: The above records are to be kept separately for each equipment.

RECORDS TO BE MAINTAINED AS PER TABLE-2 OF SIT (BY BULK PACKING UNIT)

Form No. 10

FORMAT FOR MAINTENANCE OF TEST RECORDS WEIGHMENT CONTROL AT PACKING STAGE

HOURLY CHECK OF MASS OF DRUMS

Date	Time (Hourly)	Condition of Drums	Net quantity of cement	Record of calibration of weighing scale and Date of calibration.

Form No. 11

FORMAT FOR MAINTENANCE OF RECORDS FOR THE CONDITIONS OF THE EMPTY DRUMS/BULKERS
FOR PACKING CEMENT

Date	No. of empty drums/Bulkers checked	No. of empty drums/Bulkers rejected	Reasons/Remarks	Sign of firms inspector

Form No. 12

FORMAT FOR MAINTENANCE OF TEST RECORDS WEIGHMENT CONTROL AT PACKING STAGE

HOURLY CHECK OF MASS OF BAGS

Date	Shift	Time(Hourly)	No of Bags	Net quantity of Bags from Nozzles	Records of calibration/date of calibration of nozzles

Form No. 13

RECEIPT OF CEMENTS

Date of receipt	Batch No.	Supply received from	Test Certificate No

Form No. 14
CEMENT DISPATCH DATA FROM PACKING

Date	Quantity	Passed for Standard Mark	Rejected (if any)	Reasons for not marking/Method of disposal

Form No 15 & 16
TEST DONE AT FACTORY (At receipt stage and at bulk packing terminal)

Date	Batch No.	LOI	IR	Fineness	Setting Time	Remarks

Form No 17 & 18
ORDINARY PORTLAND CEMENT (PHYSICAL TEST REPORT) (At receipt stage and at bulk packing terminal)

Date	Batch No.	Test Report	Soundness		Compressive Strength			Remarks
			LC	AC	3 days	7 days	28 days	

Note-1: Sub-contracting is permitted to a laboratory recognized by the Bureau or Government laboratories empaneled by the Bureau.

Note-2: The control unit and levels of control as decided by the Bureau are obligatory to which the licensee shall comply with.